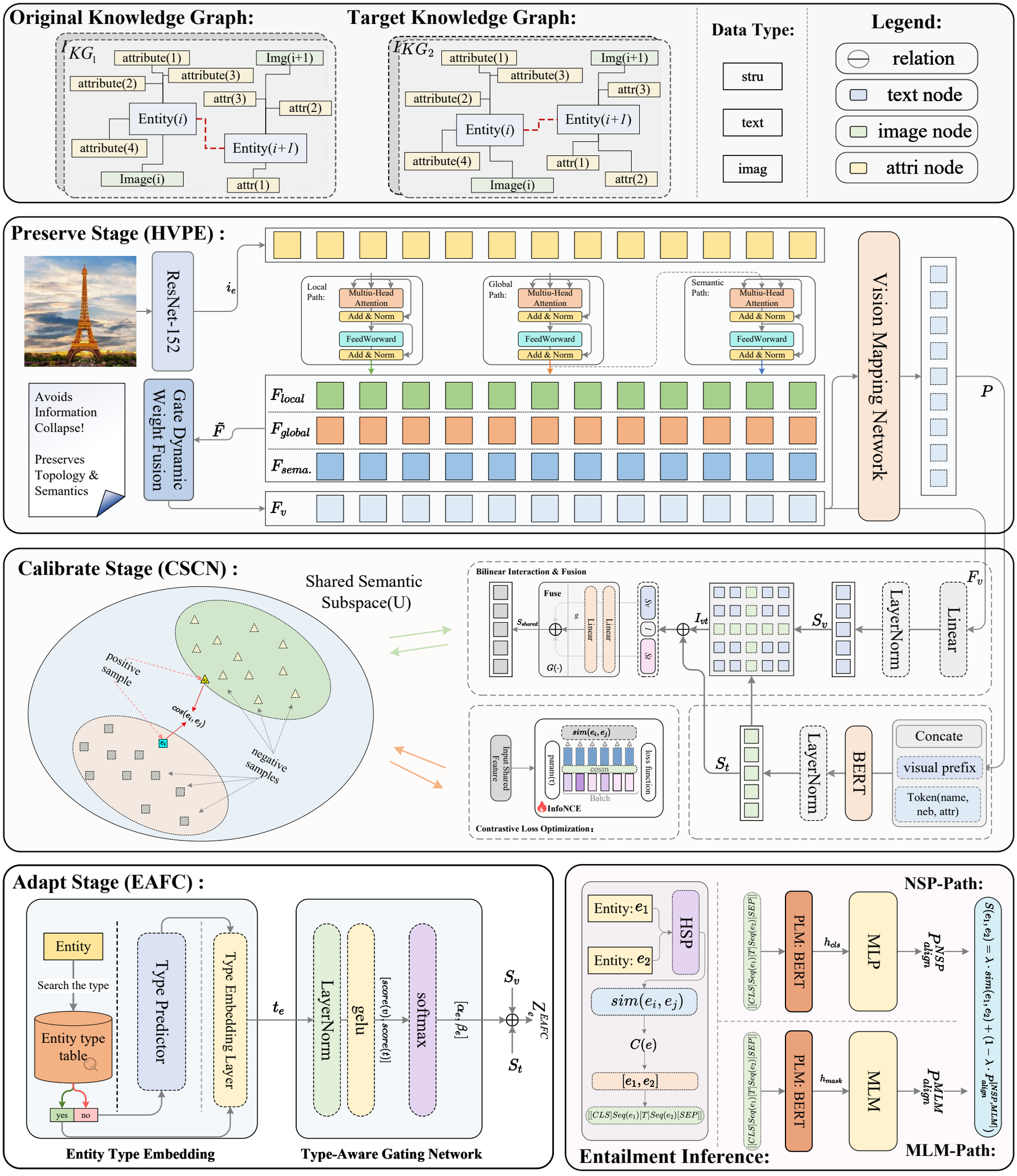
Original Knowledge Graph:
Target Knowledge Graph:
Data Type:
Legend:
relation
text node
image node
attri node
stru
text
imag
Preserve Stage (HVPE) :
ResNet-152
Local Path:
Global Path:
Semantic Path:
Multiu-Head Attention
Add & Norm
FeedWorward
Add & Norm
Vision Mapping Network
Gate Dynamic Weight Fusion
Avoids Information Collapse!
Preserves Topology & Semantics
Calibrate Stage (CSCN) :
Shared Semantic Subspace(U)
positive sample
negative samples
Bilinear Interaction & Fusion
Fuse
Linear
LayerNorm
BERT
Concate
visual prefix
Token(name, neb, attr)
Input Shared Feature
param(t)
loss function
cosin
Batch
InfoNCE
Contrastive Loss Optimization:
Adapt Stage (EAFC) :
Entity
Search the type
Entity type table
yes
no
Type Predictor
Type Embedding Layer
LayerNorm
gelu
softmax
Entity Type Embedding
Type-Aware Gating Network
Entity: e_1
Entity: e_2
HSP
PLM: BERT
MLP
MLM
NSP-Path:
MLM-Path:
Entailment Inference: